

9 (long) A polymer mixing plant includes a heater (H), an inlet valve (V) and a mixer (M). A synchronous controller system for the plant is to be designed using JK bistables. The system starts from STATE 0, where H , V , and M are turned OFF ($H = V = M = 0$). It remains in this state until a push button (B) is pressed ($B = 1$) where the system enters STATE 1, with only H turned ON. Regardless of whether B is pressed or not, the system then moves to STATE 2 where both H and M are OFF and V is ON. Again, regardless of whether B is pressed or not, the system then moved to STATE 3 where both H and M are ON and V is OFF. If B is pressed while the system is at STATE 3, it returns to STATE 2, otherwise the system returns to STATE 0.

- (a) Draw the state diagram of the mixing plant controller system. [5]
- (b) How many bistables would be needed to implement the controller? Draw the state transition table. [12]
- (c) With the aid of Karnaugh maps, develop simplified Boolean expressions for the J and K inputs of the bistables. [8]
- (d) Draw the complete circuit implementation. [5]